Semantic Search Using Word Embeddings and Approximate Nearest Neighbor Retrieval

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Unpublished working draft.
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Abstract

We propose to build and evaluate a semantic search engine that leverages word embeddings to retrieve relevant documents based on the *meaning* of a query, rather than relying on exact lexical overlap. The system encodes both documents and queries as dense vector representations using Word2Vec and BERT-based sentence encoders, indexes them with Approximate Nearest Neighbor (ANN) search structures (FAISS with HNSW), and retrieves the top-k most semantically similar documents at query time. We evaluate retrieval quality using Precision@k and Recall@k on the MS MARCO Passage Retrieval and BEIR benchmark datasets, comparing the proposed approaches against a TF-IDF baseline.

CCS Concepts

• **Information systems** → **Query representation**; *Retrieval models and ranking*; • **Computing methodologies** → Natural language processing.

Keywords

semantic search, word embeddings, Word2Vec, BERT, approximate nearest neighbor, FAISS, information retrieval, Precision@k, Recall@k

1 Motivation

Traditional information retrieval systems such as TF-IDF and BM25 rely heavily on lexical matching and often fail when there is a vocabulary mismatch between queries and documents. For example, a query such as "car engine failure" may not retrieve documents containing "automobile motor issue" despite their semantic similarity.

This limitation motivates the use of dense vector representations that capture semantic meaning. By embedding both queries and documents into a shared vector space, we enable retrieval based on meaning rather than exact word overlap.

2 Goal

The goal of this project is to design and implement a **semantic search engine** that retrieves relevant documents for a natural-language query by exploiting semantic similarity between dense vector representations, rather than relying on exact keyword matching.

The system is designed as an end-to-end pipeline consisting of embedding generation, similarity computation, efficient indexing, and retrieval.

2.1 Embedding Generation

In traditional information retrieval systems, documents and queries are represented using sparse vectors based on term frequencies. However, such representations fail to capture semantic relationships between words.

To address this, we encode both documents and queries into dense vector representations using two approaches:

- **Word2Vec (averaged embeddings):** A Word2Vec skip-gram model [3] is used to learn word-level embeddings. Each document or query is represented as the average of its constituent word vectors.
- **BERT-based sentence embeddings:** A pre-trained `sentence-transforme` model [5] encodes entire sentences or passages into fixed-dimensional vectors that capture contextual meaning.

These dense embeddings enable the system to capture semantic similarity between queries and documents even when they do not share common words.

2.2 Similarity Measure

To quantify the similarity between query and document embeddings, we use **cosine similarity**, defined as:

$$\mathrm{sim}(q, d) = \frac{q \cdot d}{||q||\,||d||} \tag{1}$$

where q and d represent the query and document vectors respectively. Cosine similarity measures the angle between vectors and is invariant to their magnitude, making it suitable for comparing embeddings.

A higher cosine similarity indicates greater semantic relevance between the query and the document.

2.3 Indexing and Retrieval

Once document embeddings are generated, they are indexed to enable efficient retrieval. We use FAISS [1] with a Hierarchical Navigable Small World (HNSW) graph structure to store and search embeddings.

At query time, the input query is encoded into a dense vector and compared against all indexed document vectors. The system retrieves the top-k nearest neighbors based on cosine similarity and returns them as ranked results.

2.4 Why Approximate Nearest Neighbor (ANN)?

Exact nearest neighbor search requires comparing the query vector with every document vector, leading to linear time complexity with respect to the number of documents. This becomes computationally expensive for large-scale datasets such as MS MARCO.

To address this, we employ **Approximate Nearest Neighbor (ANN)** search. ANN methods, such as HNSW used in FAISS, significantly reduce search time by traversing a graph-based index structure. While approximate, these methods achieve high retrieval accuracy with substantially improved efficiency.

This trade-off enables real-time semantic search over large document collections.

2.5 Research Objective

The primary research question of this project is:

To what extent does semantic (embedding-based) retrieval improve over lexical (TF-IDF/BM25) retrieval, and how do shallow (Word2Vec) and deep (BERT-based) embedding models compare under identical ANN retrieval conditions?

3 Query Flow

At query time, the system processes a user input through a sequence of steps to retrieve semantically relevant documents. The pipeline is designed to ensure consistency between document and query representations.

(1) **Query Input:** The user provides a natural-language query, which may not share exact keywords with relevant documents.
(2) **Query Encoding:** The input query is transformed into a dense vector representation using the same embedding model (Word2Vec or Sentence-BERT) used for documents. This ensures that both queries and documents lie in the same semantic vector space.
(3) **Similarity Search:** The query vector is passed to the FAISS index, where Approximate Nearest Neighbor (ANN) search is performed. The system identifies the top-k document vectors that are closest to the query vector based on cosine similarity.
(4) **Retrieval of Documents:** The indices returned by the FAISS search are mapped back to their corresponding documents in the dataset.
(5) **Ranking and Output:** The retrieved documents are ranked according to their similarity scores and presented to the user as the final search results.

This pipeline enables efficient and scalable semantic retrieval by leveraging dense embeddings and ANN-based indexing.

4 Evaluation

4.1 Datasets

MS MARCO Passage Retrieval [4]. MS MARCO provides approximately 8.8 million passages and human-annotated relevance judgments for around one million queries sourced from real-world Bing search logs. The dataset contains sparse but reliable relevance labels, making it well-suited for evaluating retrieval systems using metrics such as Precision@k and Recall@k. Due to computational constraints, we use the *dev-small* split (~6,980 queries) for rapid experimentation and the full development set for final evaluation.

BEIR Benchmark [6]. To evaluate the generalization ability of the proposed system, we use subsets of the BEIR benchmark, including NFCorpus (biomedical domain) and SciFact (scientific claim verification). These datasets differ significantly from MS MARCO in terms of domain and query distribution, allowing us to assess the robustness of embedding-based retrieval across domains.

AG News Dataset (for prototyping). In addition to the above benchmarks, we use the AG News dataset for initial development and qualitative analysis. It consists of news articles across four categories and provides a clean and manageable corpus for debugging and demonstrating semantic retrieval. Since it lacks explicit relevance labels, we simulate queries using article text.

Rationale. MS MARCO provides realistic search queries with relevance judgments, BEIR enables cross-domain evaluation, and AG News facilitates rapid prototyping and visualization. Together, these datasets provide a comprehensive evaluation framework.

4.2 Metrics

Let R_q denote the set of ground-truth relevant documents for a query q, and let $L_q^{(k)}$ denote the top-k retrieved documents.

We evaluate retrieval performance using standard information retrieval metrics:

$$\text{Precision@}k = \frac{1}{|Q|} \sum_{q \in Q} \frac{|R_q \cap L_q^{(k)}|}{k} \tag{2}$$

$$\text{Recall@}k = \frac{1}{|Q|} \sum_{q \in Q} \frac{|R_q \cap L_q^{(k)}|}{|R_q|} \tag{3}$$

Interpretation. Precision@k measures the proportion of retrieved documents that are relevant, while Recall@k measures the proportion of all relevant documents that are successfully retrieved. These metrics provide complementary insights into retrieval quality.

We evaluate at $k \in \{5, 10, 20\}$ to understand how performance varies with the number of returned results.

4.3 Baseline: BM25

BM25 is a widely used lexical retrieval model that ranks documents based on term frequency and inverse document frequency. It relies on exact keyword matching and does not capture semantic relationships between terms.

We implement BM25 using the `rank_bm25` library and use it as a baseline to compare against embedding-based retrieval methods.

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Table 1: Systems compared in the evaluation.

System	Encoder	Index
BM25 (baseline)	Sparse TF-IDF / BM25	Inverted index
Word2Vec-ANN	Avg. Word2Vec	FAISS HNSW
BERT-ANN	Sentence-BERT	FAISS HNSW

This allows us to quantify the improvement achieved by semantic search over traditional lexical approaches.

4.4 Baselines and Ablations

Table 1 summarises the evaluated systems. The BM25 baseline uses an inverted index, while both embedding-based methods use FAISS with HNSW for efficient Approximate Nearest Neighbor search.

Ablation Study. To understand the contribution of different components, we compare:

- Word2Vec vs Sentence-BERT embeddings
- ANN-based retrieval vs exact nearest neighbor search
- Performance across different values of k

In addition to accuracy metrics, we also report:

- Index construction time
- Average query latency
- Memory usage

These measurements help evaluate the practical trade-offs between retrieval quality and system efficiency.

5 Visualization

To gain insight into the structure of the learned embedding space, we visualize document representations using dimensionality reduction techniques such as Principal Component Analysis (PCA) and t-SNE.

Since embeddings typically lie in high-dimensional spaces (e.g., 384 or 768 dimensions), these techniques project them into two or three dimensions for visualization. This allows us to observe clustering behavior among semantically similar documents.

For example, documents belonging to similar topics (e.g., sports, politics, or technology) are expected to form distinct clusters in the reduced space when encoded using context-aware models such as Sentence-BERT.

Such visualizations help qualitatively evaluate the effectiveness of embedding models in capturing semantic relationships and provide an intuitive understanding of how the search system organizes information in vector space.

6 Limitations

Despite the effectiveness of embedding-based retrieval, the proposed system has several limitations:

- **Domain Sensitivity:** Pre-trained embedding models may not perform well on highly specialized domains (e.g., medical or legal text) if such data was not adequately represented during training.
- **Loss of Fine-Grained Information:** Representing an entire document as a single dense vector can lead to loss of detailed information, especially for long documents containing multiple topics.
- **Exact Match Queries:** Dense retrieval may underperform for queries that require exact keyword matching, such as searching for specific names, numbers, or identifiers.
- **Approximation Errors in ANN:** While Approximate Nearest Neighbor methods significantly improve efficiency, they may occasionally miss the exact nearest neighbors, leading to minor reductions in retrieval accuracy.

Understanding these limitations is important for interpreting system performance and motivates future improvements such as hybrid retrieval and re-ranking techniques.

7 Related Work

Recent advances in information retrieval have shifted from traditional lexical methods to dense retrieval techniques that leverage neural representations of text.

Early approaches such as TF-IDF and BM25 rely on exact term matching and fail to capture semantic similarity between queries and documents. To address this limitation, Word2Vec [3] introduced distributed word representations that encode semantic relationships based on contextual co-occurrence.

Building on this idea, BERT-based models [5] and bi-encoder architectures [2] enable encoding of entire sentences or passages into dense vectors. These models have shown significant improvements in semantic search and question-answering tasks.

Efficient retrieval in dense vector spaces is made possible by Approximate Nearest Neighbor (ANN) algorithms, such as those implemented in FAISS [1]. Graph-based methods like HNSW provide fast and scalable search with minimal loss in accuracy.

Our work focuses on the *evaluation and comparison* of shallow (Word2Vec) and deep (BERT-based) embedding methods under a unified ANN retrieval framework, using established IR benchmarks.

8 Timeline

Week 1 Dataset preparation, preprocessing, and implementation of Word2Vec embeddings.

Week 2 Integration of Sentence-BERT embeddings and construction of FAISS index for retrieval.

Week 3 Evaluation on MS MARCO and BEIR datasets, including computation of Precision@k and Recall@k, along with baseline comparison (BM25).

Week 4 Development of Streamlit demo interface, analysis of results, and final report preparation.

Acknowledgments

We would like to thank the instructors and teaching assistants of the CS6200 Information Retrieval course at IIT Gandhinagar for their guidance and valuable feedback throughout the project.

We also acknowledge the creators of the MS MARCO and BEIR datasets for providing high-quality benchmark datasets, and the developers of open-source libraries such as FAISS, SentenceTransformers, and Gensim that made this project feasible.

Additionally, we thank the research community for foundational work in semantic search, word embeddings, and dense retrieval that inspired this project.

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References

[1] Jeff Johnson, Matthijs Douze, and Hervé Jégou. 2019. Billion-scale similarity search with GPUs. *IEEE Transactions on Big Data* 7, 3 (2019), 535–547.

[2] Vladimir Karpukhin, Barlas Oguz, Sewon Min, Patrick Lewis, Ledell Wu, Sergey Edunov, Danqi Chen, and Wen-tau Yih. 2020. Dense Passage Retrieval for Open-Domain Question Answering. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. 6769–6781.

[3] Tomas Mikolov, Kai Chen, Greg Corrado, and Jeffrey Dean. 2013. Efficient Estimation of Word Representations in Vector Space. *arXiv preprint arXiv:1301.3781* (2013).

[4] Tri Nguyen, Mir Rosenberg, Xia Song, Jianfeng Gao, Saurabh Tiwary, Rangan Majumder, and Li Deng. 2016. MS MARCO: A Human Generated Machine Reading Comprehension Dataset. In *Proceedings of the Workshop on Cognitive Computation: Integrating Neural and Symbolic Approaches*.

[5] Nils Reimers and Iryna Gurevych. 2019. Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP)*.

[6] Nandan Thakur, Nils Reimers, Johannes Daxenberger, and Iryna Gurevych. 2021. BEIR: A Heterogeneous Benchmark for Zero-shot Evaluation of Information Retrieval Models. In *Proceedings of the Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*.

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